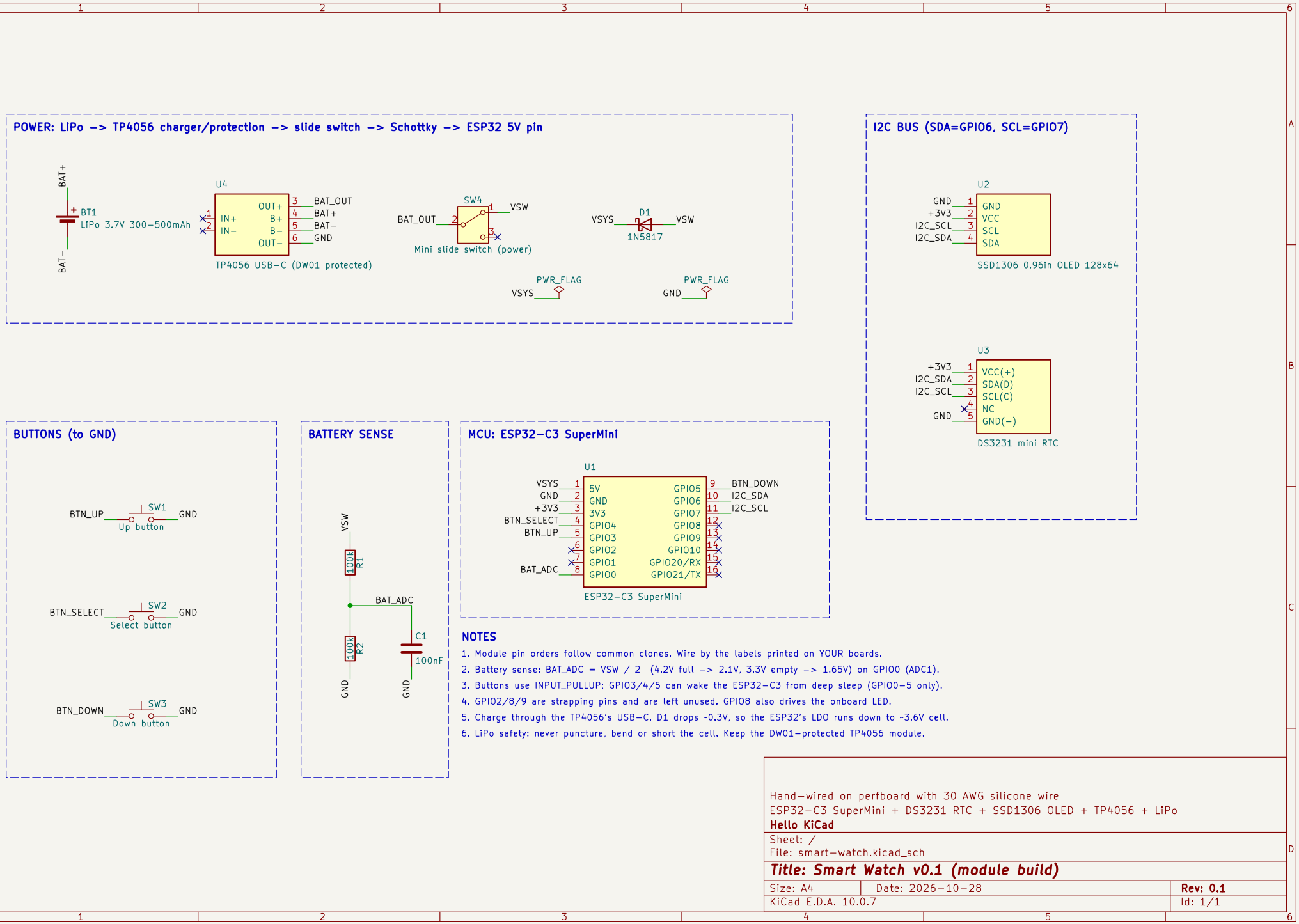


NOTES

1. Module pin orders follow common clones. Wire by the labels printed on YOUR boards.
2. Battery sense: BAT_ADC = VSW / 2 (4.2V full -> 2.1V, 3.3V empty -> 1.65V) on GPIO0 (ADC1).
3. Buttons use INPUT_PULLUP; GPIO3/4/5 can wake the ESP32-C3 from deep sleep (GPIO0-5 only).
4. GPIO2/8/9 are strapping pins and are left unused. GPIO8 also drives the onboard LED.
5. Charge through the TP4056's USB-C. D1 drops ~0.3V, so the ESP32's LDO runs down to ~3.6V cell.
6. LiPo safety: never puncture, bend or short the cell. Keep the DW01-protected TP4056 module.

Hand-wired on perfboard with 30 AWG silicone wire
ESP32-C3 SuperMini + DS3231 RTC + SSD1306 OLED + TP4056 + LiPo
Hello KiCad

Sheet: /
File: smart-watch.kicad_sch

Title: Smart Watch v0.1 (module build)

Size: A4	Date: 2026-10-28	**Rev: 0.1**
KiCad E.D.A. 10.0.7		Id: 1/1